Leema Krishna Murali

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Education

University at Buffalo, NY **2021**
Master of Science - Biomedical Engineering (GPA: 3.75/4.0)
Areas of research: AI in medical imaging, computer vision, machine learning, and deep learning
Master Thesis - Multi-Compartment Segmentation in Renal Transplant Pathology [8]

Ramaiah Institute of Technology, Bangalore, India **2016**
Bachelor of Engineering – Medical Electronics (CGPA: 9.37/10.0)
Best undergraduate research project for the year 2016 in the Department [10]

Core Technical Skills

- **Programming Languages/Libraries:** Python, MATLAB, C++, Java, JavaScript, Apache Cordova, R programming, Shiny, SQL, PyTorch, Tensorflow, OpenCV, NumPy, SciPy, Pandas, Scikit-Learn, Linux shell scripting.
- **Tools/Software:** 3D Slicer, DICOM, ANTS, SPM, ImageJ, QuPath, Aperio, FreeSurfer, CUDA, GIT
- **Cloud computing & containerization tools:** AWS, Databricks, Azure, GCP, Hadoop, PySpark, Databricks, MLflow, Docker
- **Keywords:** Machine Learning, Deep Learning, Computer Vision, Foundational AI, Natural Language Processing, Multimodal Machine Learning, Self-supervised Learning, Large Language Models, Vision Language Models, Robust Training, Data/Parameter Efficient Training, Medical Imaging

Research & Industry Experience

Senior Data Scientist **(10/2020 – Present)**
Eisai Pharmaceuticals, New Jersey

- Identified novel biomarkers in early therapy areas using image segmentation, registration, and interpretable machine learning techniques by curating and combining diverse information from complex unstructured and time-series structured data from clinical trials for disease identification and progression models.
- Developed, implemented, and evaluated patch-based and voxel-based deep neural network architectures using FCN and CNN on MRI scans to generate dense local predictions of amyloid and tau accumulation in the brain.
- Implemented and evaluated LLMs to evaluate EHR data from real-world medical and clinical trial data for disease stage identification in early and late Alzheimer's condition.
- Spearheaded a companywide cloud-based pipeline for high-throughput, robust, and fast processing of neuroimaging scans reducing the cost by million dollars.
- Developed multi-modal models using pre-trained graph-GAN (transfer learning), RNN, LSTM, gradient boosting, and diffusion models for predictive modeling of Alzheimer's disease progression and drug response using real-world and clinical trial data achieving a sensitivity of 81% and specificity of 71%, enhancing patient selection.
- Mentored cross-functional teams on ML/DL algorithms and their research in medicine.

AI Research Fellow – Open Science Research **(08/2025 – Present)**
Cohere Labs Community, Remote

- Investigated several state-of-the-art VLMs on a set of spatial reasoning benchmarks to understand the performance gaps in capturing geometric relations and spatial layouts.
- Implemented 2D Rotary Positional Encoding in image encoder (like SigLIP, CLIP, SigLIP2, and AIMv2)-LLM alignment within the LLaVA framework to demonstrate their role in improving the preservation of 2D spatial information and enhancing the spatial reasoning performance of VLMs.

MedARC (Medical AI Research Center), Remote

- Contributing to open-source medical AI research projects, like pathology [GitHub Repo] and fMRI [GitHub Repo] foundation models.
- Contributed a key engineering solution to improve the pathology foundation model's architecture by implementing extended field of view sampling and dynamic cropping, eliminating the giga-pixel bottleneck inherent in whole slide image processing.
- Implemented a preprocessing pipeline like fMRIPrep of raw 4D volumetric fMRI data for fMRI foundation model input representation that includes steps like co-registration, segmentation, normalization, and skull-stripping.
- (Ongoing) Refining DINOv3 self-supervised objective of pathology foundation model against collapse failure modes to achieve high-quality and generalizable feature sets.

Research Assistant, SUNY Research Foundation **(11/2018 – 08/2020)**
University at Buffalo, New York

- Developed adversarial DL models (inspired by GAN) and investigated the histological performance of generative models using high-performance clusters (HPC) in modeling renal glomerular structure.
- Implemented supervised DL-based domain translation and stain normalization of real and AI-powered generated data comprising routinely stained (PAS) brightfield whole slide images of kidney biopsy sections.
- Developed DL-based object detection models (inspired by GAN) for multi-compartments from multiplex immunohistochemical (IHC) images in renal pathology and investigated their performance over U-Net models.

Bosch Healthcare **(07/2016 – 07/2018)**
Associate Software Engineer, India

- Developed an enhanced version of a HIPAA-compliant telemedicine application (front-end and back-end web and mobile app development) deployed in AWS as a pilot study to connect doctors and patients through video conferencing.
- Expedited product evaluation and market study by designing and testing sensor-based prototypes with Bluetooth communication protocol integrated with an Android application for real-time monitoring and participated in various customer forums and conferences to demonstrate the company's healthcare solutions.

Research Assistant, Center for Medical Electronics and Computing **(01/2015 – 06/2016)**
Ramaiah Institute of Technology, India

- Developed and tested an image processing algorithm that used techniques like contrast-limited adaptive histogram equalization (CLAHE), and color-thresholding.
- Implemented semi-automated detection of ROI and extracted features like statistical, and textual features. The novelty of this study was the extraction of a new feature named "Nipple Asymmetry". The classification accuracy of the dataset of 20 thermal images using the KNN Classifier is 75% using nipple asymmetry.

Publications

1) Alam, Nahid, **Murali, Leema Krishna**, et al. "Spatial Reasoning is Not a Free Lunch: A Controlled Stud on LLaVA?", **ICLR 2026 Workshop on I Can't Believe It's Not Better: Where Large Language Models Need to Improve, Rio de Janeiro, Brazil.**
2) Zafar Anas*, **Murali, Leema Krishna***, Vashist, Ashish, "Beyond Accuracy: Evaluating Visual Grounding in Multimodal Medical Reasoning", **ICLR 2026 Workshop on Catch, Adapt, and Operate: Monitoring ML Models Under Drift, Rio de Janeiro, Brazil.** *Equal Contribution
3) Devanarayan, Viswanath, Charil, Arnaud, Horie, Kanta, Doherty, Thomas, Llano Daniel A, Andreozzi, Erica, Sachdev, Pallavi, Ye, Yuanqing, **Murali, Leema Krishna**, et al. "Plasma pTau217 ratio predicts continuous regional brain tau accumulation in amyloid-positive early Alzheimer's disease." ***Alzheimer's & Dementia*** **21.2 (2025)**: e14411. (DOI: 10.1002/alz.14411) (Peer-reviewed published)
4) **Murali, Leema**, et al. "Volumes of specific substructures within the Amygdala and Hippocampus are impacted by brain amyloid-β." ***Alzheimer's Association International Conference.*** **ALZ, 2024**. (Poster presentation)
5) Devanarayan, Viswanath, Doherty, Thomas, Charil, Arnaud, Sachdev, Pallavi, Ye, Yuanqing, **Murali, Leema Krishna**, et al. "Plasma pTau217 predicts continuous brain amyloid levels in preclinical and early Alzheimer's disease." ***Alzheimer's & Dementia*** **20.8 (2024)**: 5617-5628. (DOI: 10.1002/alz.14073) (Peer-reviewed published)
6) Charil, Arnaud, Devanarayan, Viswanath, **Murali, Leema**, et al. "Baseline regional Tau distribution predicts fast cognitive decline in subjects with mild cognitive impairment." ***Alzheimer's & Dementia*** **18 (2022)**: e068154. (DOI: 10.1002/alz.068154) (Peer-reviewed abstract)
7) Devanarayan, Viswanath, Charil, Arnaud, Nelson, Todd, Qi, Xin, **Murali, Leema**, et al. "Development and validation of AI-based tools for brain amyloid-β detection using MRI." ***Alzheimer's & Dementia*** **18 (2022)**: e068231. (DOI: 10.1002/alz.068231) (Peer-reviewed abstract)
8) Lutnick, Brendon, **Murali, Leema Krishna**, et al. "Histo-fetch–On-the-fly processing of gigapixel whole slide images simplifies and speeds neural network training." ***Journal of Pathology Informatics*** **13 (2022)**: 100169. (DOI: 10.4103/jpi.jpi_59_20) (Peer-reviewed published)
9) Jen, Kuang-Yu*, **Murali, Leema Krishna***, et al. "In Silico Multi-Compartment Detection Based on Multiplex Immunohistochemical Staining in Renal Pathology." ***Medical Imaging 2021: Digital Pathology***. Vol. 11603. SPIE, 2021. (DOI: 10.1117/12.2581795) (Poster presentation and peer-reviewed abstract) *Equal Contribution
10) **Murali, Leema Krishna**, et al. "Generative modeling for renal microanatomy." ***Medical Imaging 2020: Digital Pathology***. Vol. 11320. SPIE, 2020. (DOI: 10.1117/12.2549891) (Conference presentation and proceedings paper)
11) Natarajan, Sriraam*, **Murali, Leema***, et al. "Classification of Breast Thermograms Using Statistical Moments and Entropy Features with Probabilistic Neural Networks." ***International Journal of Biomedical and Clinical Engineering (IJBCE)*** **6.2 (2017)**: 18-32. (Peer-reviewed published) *Equal Contribution

Honors & Awards

- FY2021 Eisai Innovation Paper Contest Award, Eisai Inc. (**2022)**
- Neurology Clinical R&D Breaking through award for being relentless and innovative, Eisai Inc. (**2021)**
- Gold Medalist and First rank honors award in B.E. in Medical Electronics, M.S. Ramaiah Institute of Technology (**2016**)
- 2nd Best research project award at "Pradarshana-2016" Project exhibition, M.S. Ramaiah Institute of Technology (**2016**)
- Certificate of Merit for Academic Excellence (12th Grade), CMR National PU College (**2012**)
- Vidhya Madhvi Commemoration Prize for Resilience and Academic Achievement (Grade 7), St. Francis Xavier Girls' High School (**2007**)

Volunteering & Teaching Experience

- ML-Healthcare Community Lead and Researcher at Cohere Labs Community (**2025**)
- Student volunteer at Science is Elementary at SUNY Buffalo, leading students through hands-on STEM activities at Westminster Charter School. (**2018-2020**)
- Volunteer and member of Society of Women Engineers (SWE), mentored fellow women students in data science and AI in medical imaging. (**2019**)
- Volunteer at Grace Hopper Celebration – volunteered to manage and monitor technical talks in the field of AI at the Grace Hopper Conference. (**2020**)
- Volunteer and member of National Service Scheme – volunteered to organize blood donation camps and teach science and soft skills to students in public schools. (**2014-2016**)

Professional Development and Certifications

- Mathematics for Machine Learning Specialization, Imperial College London 2023
- Generative AI with Transformers, DeepLearning.AI 2024
- AI Safety and Alignment, Rapid Upskilling Cohort, AISI, Georgia Tech 2025